

Meitner discussed the matter with her nephew, Otto Frisch, a physicist in Denmark. Applying Einstein's theory of relativity, she realized that the mass was not lost, but was converted into energy. When the nuclei of the uranium atoms divided (nuclear fission), forming two new nuclei that together weighed less than the original uranium nucleus, an enormous amount of energy was created. Frisch repeated the results in his Copenhagen laboratory, and Meitner published their joint findings on the nuclear fission of uranium in early 1939. News of the discovery spread rapidly and ultimately led to the US military creating an atomic bomb (see box).

The Nobel Prize for the discovery of nuclear fission was awarded to Otto Hahn in 1944—Meitner's role went unrecognized for decades. However, she did not seem bitter about her omission, saying that "Science can bring both joy and satisfaction to your life."

"Science makes people reach selflessly for **truth and objectivity."**

Lise Meitner, 1953

In Hahn's 1938 experiment, Meitner explained the process of nuclear fission, which occurs when a nucleus is bombarded with a neutron and then splits, releasing a vast amount of energy.

